

6.2 The RAM control law

6.2.1 The dual control law for RAM modes

The formulation of the dual control law, used in RAM, is an improved version of that used before in the IRAS satellite (see the next chapter on ROM control for the old law). Its advantage over the previous version lies in a constant velocity gain over the whole working range. In the current version of the law, the optimal slew behaviour is obtained by *progressively reducing the attitude feedback gain*, instead of by increasing the stiffness of the velocity feedback, as in the old version. This new formulation is:

$$T_{req} = P \cdot \frac{e}{\sqrt{1 + R \cdot |e|}} + D \cdot \dot{e} \quad (6-2)$$

$$R = \frac{P^2}{D^2} \cdot \frac{1}{2 \cdot \alpha_0} \quad (6-3)$$

It is obvious by inspection, that in the vicinity of the origin, while $R \cdot |e| \ll 1$, the law is a conventional PD law. For large errors and velocities, it defines a constant-deceleration parabola at zero torque request. The torque requests to one side of this parabola are positive; to the other side, they are negative:

$$e = -\frac{\dot{e}^2}{2 \cdot \alpha_0} \quad (6-4)$$

The reason is, that for $R \cdot |e| \gg 1$, the feedback reduces to:

$$T_{req} \approx D \cdot \sqrt{2 \alpha_0 \cdot e} + D \cdot \dot{e} \quad (6-5)$$

Using equation (6-3) we can rewrite the locally linearised control gains to:

$$\begin{aligned} \frac{dT_{req}}{de} &\rightarrow \frac{P}{2\sqrt{R}} \cdot \frac{1}{\sqrt{|e|}} \\ \frac{dT_{req}}{d\dot{e}} &= D \end{aligned} \quad (6-6)$$

The velocity gain is the same everywhere. The attitude error gain is inversely proportional to the square root of the distance to target, and hence tends to zero for very large control errors. A useful diagram for visualizing the action of this law is the 'phase-plane plot'. Figure 6.1 at the end of the text shows the parabolic shape of the dual law (with the torque limited to a maximum) in such a diagram.

6.2.2 Integrated velocity controller

The angular velocity of the ISO satellite is constrained to a maximum value. This is done in order to prevent any wheel overloading and undue inertial effects. This restriction is implemented in a very simple manner in the dual law itself, by limiting the control error which is fed to the control law to a corresponding maximum. The phase plane plot will show that limiting the error is preferable to limiting the velocity, which might seem more obvious at first sight. The appropriate value of $e_{\max}$ is found by substituting $T = 0$ and $\dot{e} = \dot{e}_{\max}$ into the control law:

$$E_{acc} = \frac{(\dot{e}_{\max})^2}{2 \cdot \alpha_0} \quad (6-7)$$

$$E_{\max} = E_{acc} \left(\frac{1}{2} + \frac{1}{2} \sqrt{1 + \frac{4}{E_{acc} \cdot R}} \right) \quad (6-8)$$

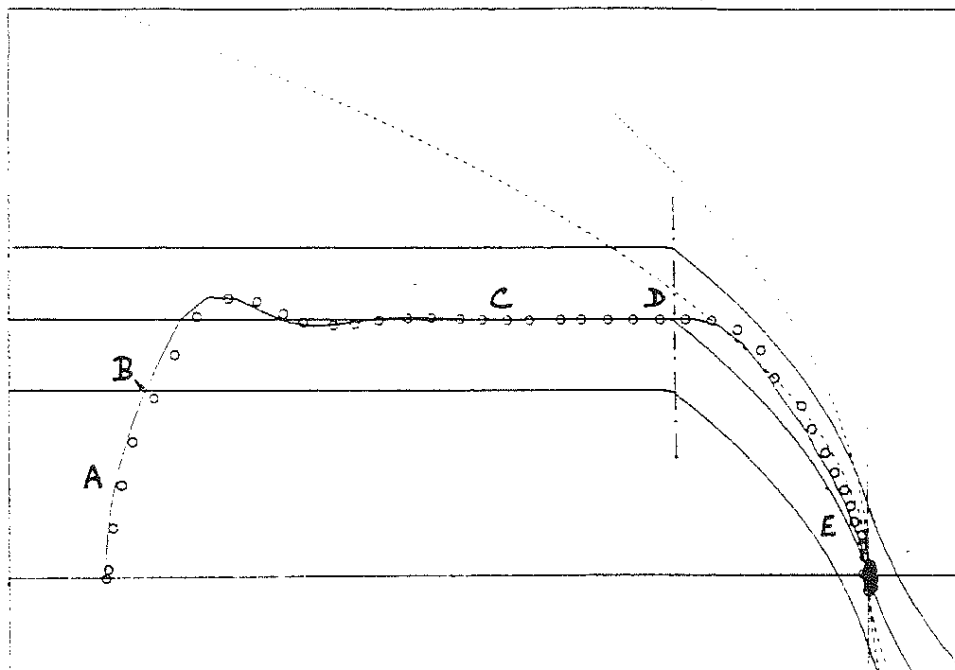
The dual law algorithm including this additional velocity controller is shown in the box below. Figure 6.1 at the end of the text give a phase plane sketch. This shows how this integrated velocity controller gives a very smooth and natural transition between velocity control and the dual law proper.

$$e_{lim} = \text{sign}(e) \cdot \min(|e|, E_{\max})$$

$$T_{req} = P \cdot \frac{e_{lim}}{\sqrt{1 + R \cdot |e_{lim}|}} + D \cdot e$$

$$R = \frac{P^2}{D^2} \cdot \frac{1}{2 \cdot \alpha_0}$$

(6-9)



*phase plane plot of a 10 arcminute slew,
with 40 arcsec/s velocity limit*

phase		torque
A	acceleration	maximum available torque
B	switch to velocity controller; accelerate to maximum speed	torque gradually reducing to zero
C	coast at maximum velocity	zero torque
D	switch to dual law, decelerate with nominal deceleration	maximum guaranteed torque
E	gradual transition to pointing on PD law	torque reducing to zero

figure 6.4: Phase plane sketch of a typical RAM slew.